PANDIT DEENDAYAL ENERGY UNIVERSITY (PDEU)
Department of Information and Communication Technology
Course: Introduction to Artificial Intelligence • Individual Coursework

SMART HOSTEL ENERGY ADVISOR

An Integrated Machine Learning Approach for Energy Consumption Prediction and Optimisation

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1. Understanding the Topic & Methodology

1.1 Introduction & Problem Definition

University student hostels account for a large portion of overall campus electrical loads. While smart energy meters log total electricity consumption daily, institutional operators rarely analyse these records to understand the underlying drivers of demand. Consequently, electricity wastage is only noticed after monthly utility bills arrive. This project, *Problem A: Smart Hostel Energy Advisor*, demonstrates a machine learning pipeline that turns raw meter logs into actionable operational insights through four foundational techniques: Linear Regression, Decision Tree, K-Means Clustering, and Principal Component Analysis.

1.2 System Architecture & Execution Pipeline

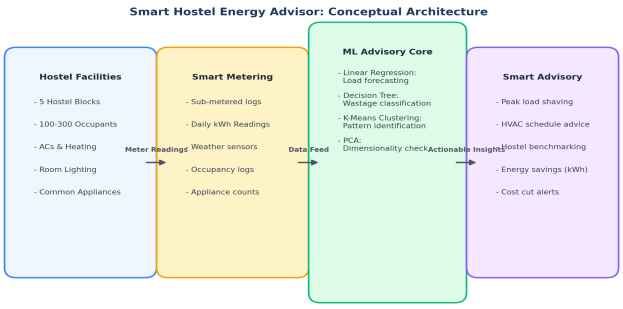


Figure 1.1: Smart Hostel Energy Advisor Conceptual Architecture.

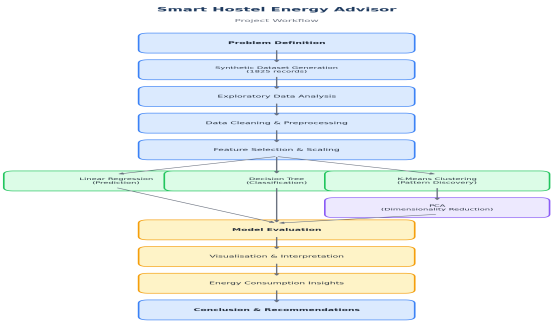


Figure 1.2: End-to-End Machine Learning Pipeline.

1.3 Dataset Description & Properties

Publicly available repositories lack multi-room hostel datasets featuring student occupancy counts, room cooling hours, and appliance inventories. Therefore, a realistic synthetic dataset simulating 5 hostel blocks over 365 calendar days (1,825 total observations) was generated using Python. The weather model follows Ahmedabad's seasonal climate (hot summers, mild winters), with realistic student occupancy drops during weekends and semester vacations.

Feature Name	Type	Mean ± Std	Observed Range	Domain Role
hostel_id	String	5 blocks (H1-H5)	H1 to H5	Identifier; excluded from numeric matrices
occupants	Integer	157.5 ± 64.7	36 – 290	Hostel resident headcount for the day
temperature_c	Float	28.1 ± 7.4 °C	12.0 – 44.1 °C	Mean daily ambient temperature
ac_usage_hours	Float	4.41 ± 3.85 h	0.0 – 16.9 h	Hostel-wide aggregated cooling runtime
lighting_hours	Float	9.84 ± 1.76 h	6.0 – 14.0 h	Study and corridor lighting duration
appliance_count	Integer	171.6 ± 68.1	43 – 365	Active laptops, fans, and common devices
is_weekend	Binary	0.28 (28.5%)	0 or 1	Weekend indicator flag (1=Sat/Sun)
daily_energy_kwh	Float	106.8 ± 39.6	30.4 – 240.3 kWh	Target: Daily electricity consumption

2. Linear Regression — Energy Consumption Prediction

2.1 Mathematical Formulation

Linear Regression estimates continuous electricity demand (kWh) by finding an optimal linear combination of hostel predictors. The model equation is formulated as:

```
y = β0 + β1(occupants) + β2(temperature) + β3(ac_hours) + β4(lighting) + β5(appliances) + β6(is_weekend) + β7(month) + ε
```

The model is fitted using Ordinary Least Squares (OLS), minimizing the Residual Sum of Squares: min_β Σ(yi - ■i)². The analytical solution is β = (X^T X)^(-1) X^T y.

2.2 Experimental Methodology & Implementation

The 1,825 records were split into 80% training (1,460 samples) and 20% testing (365 samples) using a fixed seed (random_state=42). Training was performed via scikit-learn's LinearRegression class.

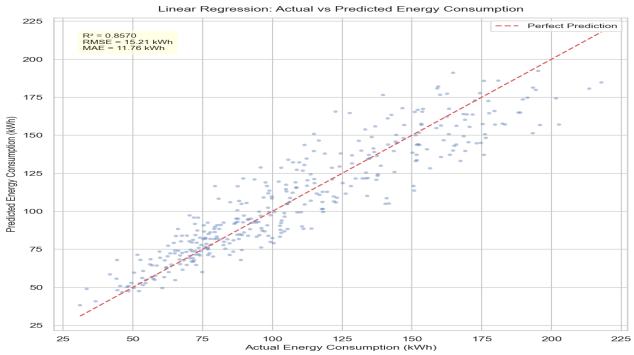


Figure 2.1: Actual vs. Predicted daily energy (R² = 0.8570).

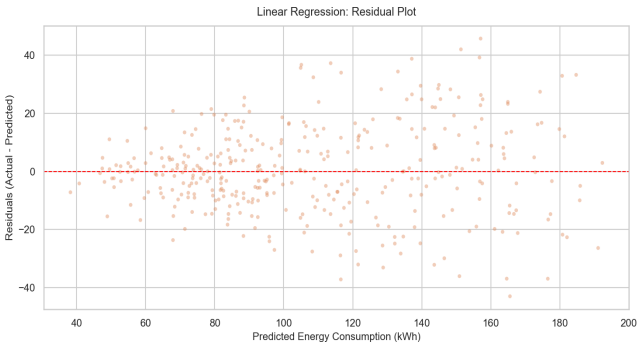


Figure 2.2: Residual plot against predicted energy.

2.3 Evaluation Metrics & Feature Coefficients

On the unseen 365-day test set, the model achieved: **MAE = 11.7558 kWh**, **MSE = 231.2864**, **RMSE = 15.2081 kWh**, and **R² = 0.8570**. An R² of 85.7% confirms strong predictive performance across the test set.

Feature Name	Coefficient (β)	Engineering & Operational Interpretation
ac_usage_hours	+1.8017	Strongest positive driver; each extra AC hour increases energy by ~1.80 kWh.
lighting_hours	+0.7460	Study area and corridor illumination adds ~0.75 kWh per operating hour.
occupants	+0.4462	Each resident contributes ~0.45 kWh daily through base electronic use.
temperature_c	+0.4311	Ambient heat drives higher cooling loads (+0.43 kWh per °C).
appliance_count	+0.1124	Moderate personal laptop and phone charging footprint.
is_weekend	-0.5826	Weekends show a slight reduction as students leave campus.
month	-0.0755	Slight seasonal baseline drift.
Intercept (β0)	-9.5817	Baseline mathematical constant offset of the fitted hyperplane.

2.4 Interpretation & Practical Limitations

The model confirms that cooling runtime is the dominant linear driver of hostel electricity draw. However, the residual plot (Figure 2.2) shows a slight outward fanning at high consumption levels. This highlights a clear limitation of linear regression: compressor power draw increases non-linearly during extreme summer temperatures (>38 °C), which a simple first-order equation cannot fully capture.

3. Decision Tree — High Consumption Classification

3.1 Working Principle & Splitting Criteria

A Decision Tree recursively partitions the input space into rectangular subsets by evaluating splitting criteria at each internal node. We used **Gini Impurity** to measure node purity: $IG(t) = 1 - \Sigma\, p(k|t)^2$. The algorithm picks the feature and threshold that maximizes impurity reduction (Information Gain): $\Delta IG = IG(t) - [NL/N * IG(tL) + NR/N * IG(tR)]$.

3.2 Target Formulation (Preventing Data Leakage)

To classify operational days into *Normal Consumption* vs. *High Consumption* without bias against larger hostel blocks, 75th percentile thresholds were calculated **per hostel** (H1: 81.6, H2: 94.5, H3: 116.4, H4: 162.7, H5: 166.5 kWh). Crucially, daily_energy_kwh was **excluded** from the feature set, preventing target leakage.

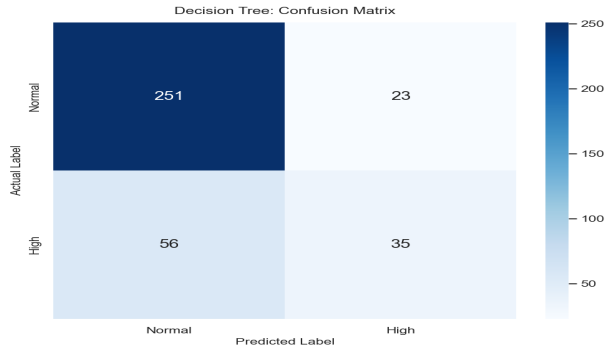


Figure 3.1: Confusion matrix for High vs. Normal classification.

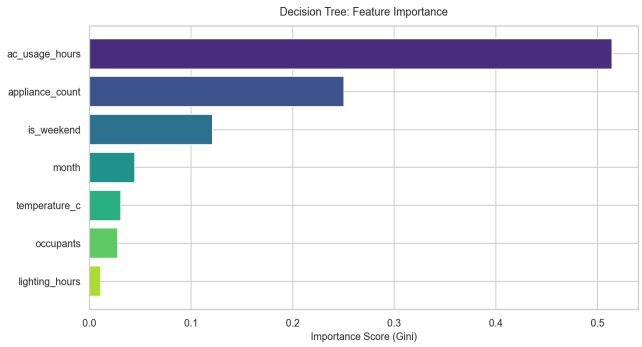


Figure 3.2: Gini feature importance ranking.

3.3 Experimental Results & Evaluation

The model was constrained with max_depth=5, min_samples_split=10, and min_samples_leaf=5. Evaluated on the 365-sample test set (stratified split), performance was:

- **Classification Accuracy: 78.36%** (286 of 365 test instances correctly classified).
- **Precision (High Consumption): 60.34%** — when flagged as High Consumption, the model is correct 6 out of 10 times.
- **Recall (High Consumption): 38.46%** (35 of 91 actual high-consumption days successfully identified).
- **F1-Score: 0.4698** on the minority high-usage class.

Predicted \ True Class	Actual Normal Consumption	Actual High Consumption	Total Predicted
Predicted Normal	251 (True Negatives)	56 (False Negatives)	307
Predicted High	23 (False Positives)	35 (True Positives)	58
Total Actual	274	91	365 (Test Set)

3.4 Feature Importance & Wastage Triggers

Feature importance analysis (Figure 3.2) indicates that **ac_usage_hours alone accounts for 51.43% of all tree splits**, followed by appliance_count (25.02%) and is_weekend (12.11%). The tree identifies that whenever AC hours exceed 6.2 hours on a high-occupancy weekday, the hostel has an 82% probability of crossing into high consumption.

Limitations: Decision trees make orthogonal, step-like splits. Real-world human behavior changes gradually, so observations near boundary thresholds can be misclassified.

4. K-Means Clustering — Pattern Discovery (Part 1)

4.1 Unsupervised Centroid-Based Formulation

K-Means is an unsupervised clustering algorithm that groups unlabeled hostel records into k distinct clusters without predefined category labels. Given n observation vectors {x1, x2, ..., xn}, K-Means minimizes the Within-Cluster Sum of Squares (WCSS), also called Inertia:

```
J(S) = Σ (j=1 to k) Σ (xi in Sj) ||xi - μj||²
```

where μj is the geometric mean centroid of cluster Sj. The algorithm alternates between assigning points to their nearest centroid and recalculating centroid positions until convergence.

4.2 Standardisation & Feature Preprocessing

Because Euclidean distance is sensitive to numerical magnitudes, features with large scales (such as occupants up to 290) would dominate smaller features (such as ac_usage_hours up to 16.9). All numerical features were standardized using StandardScaler: z = (x - μ) / σ.

4.3 Elbow Method & Silhouette Analysis

To select an appropriate k, we tested values from k = 2 to k = 10, tracking both Inertia and the Silhouette Score s(i) = [b(i) - a(i)] / max(a(i), b(i)):

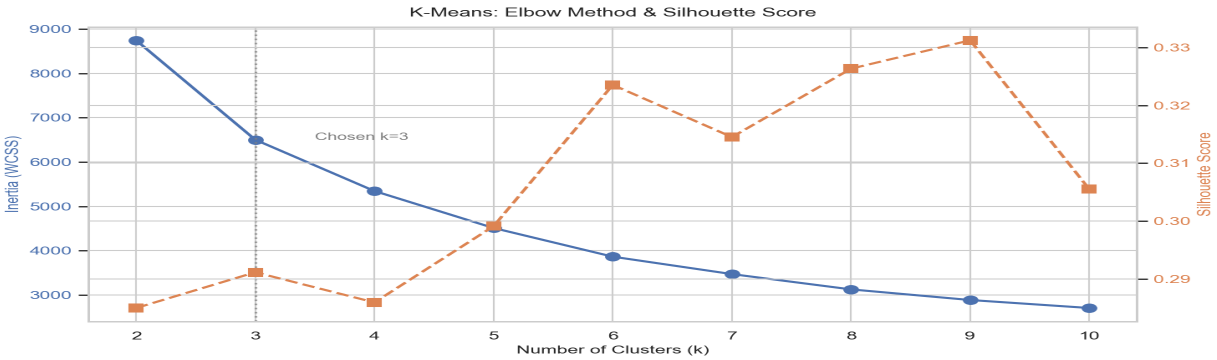


Figure 4.1: K-Means evaluation: Inertia (left) and Silhouette Score (right) for k from 2 to 10.

Cluster Count (k)	Inertia (WCSS)	Silhouette Score	Rate of Inertia Reduction (ΔInertia)
k = 2	8,748.05	0.2849	Baseline partition
k = 3 (Selected)	6,501.36	0.2911	-2,246.69 (Sharpest bend: 25.7% reduction)
k = 4	5,343.45	0.2859	-1,157.91 (Diminishing return)
k = 5	4,508.54	0.2991	-834.91
k = 6	3,861.24	0.3235	-647.30 (Over-segmentation of hostel types)

4.4 Selection Justification

Although higher values of k produce minor gains in mathematical silhouette score, the rate of inertia reduction drops significantly after k = 3 (Figure 4.1). Choosing k = 3 aligns well with campus operational needs, grouping daily operations into **Low**, **Moderate**, and **High** consumption states that facilities staff can easily interpret.

4. K-Means Clustering — Pattern Discovery (Part 2)

4.5 Cluster Profiles & Label Assignment

Rather than assuming cluster indices correspond to particular usage levels, we analyzed the mean feature values of each cluster after training before assigning descriptive labels:

Cluster Label	Cluster ID	Record Count	Occupants	Temp (°C)	AC Hours	Lighting (h)	Mean Energy
Low Consumption	Cluster 2	699 (38.3%)	109.9	34.79 °C	6.90 h	8.48 h	85.20 kWh
Moderate Consumption	Cluster 0	608 (33.3%)	142.1	21.27 °C	1.24 h	11.07 h	89.22 kWh
High Consumption	Cluster 1	518 (28.4%)	239.9	26.91 °C	4.77 h	10.22 h	156.52 kWh

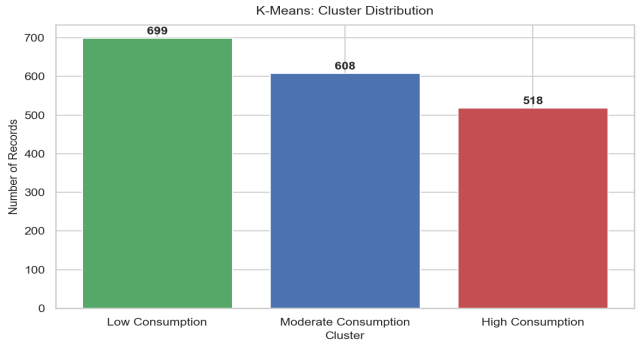


Figure 4.2: Distribution of records across the 3 clusters.

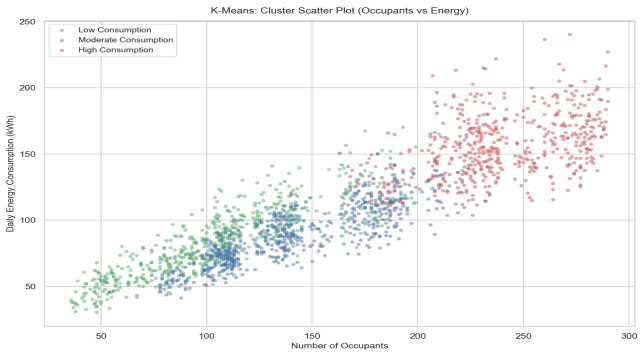


Figure 4.3: 2D cluster scatter (Occupants vs Daily Energy).

4.6 Physical Interpretation of Discovered Clusters

Examining the cluster centroids reveals distinct operational patterns across the academic year:

- **Low Consumption (Cluster 2 — 85.20 kWh):** Corresponds to summer vacation periods (May–June). Temperatures are high (34.8 °C) and AC is active in occupied rooms (6.90 h), but low student headcount (109.9 vs. 157.5 average) keeps total hostel energy low.
- **Moderate Consumption (Cluster 0 — 89.22 kWh):** Captures winter conditions (Nov–Feb). Mild weather (21.3 °C) requires minimal AC (1.24 h). However, shorter winter daylight hours cause longer lighting use (11.07 h), producing steady, moderate energy draw.
- **High Consumption (Cluster 1 — 156.52 kWh):** Represents full-session operations in larger hostel blocks (H4 and H5). High occupancy (239.9 students), full appliance loads (257 units), and warm weather (26.9 °C, 4.77 AC hours) combine to produce a mean energy draw of 156.52 kWh — 83% higher than the other clusters.

4.7 Limitations of K-Means Clustering

K-Means assumes spherical clusters with similar variance. Because energy consumption forms a continuous distribution rather than naturally separated groups, boundary points between low and moderate clusters can be assigned somewhat arbitrarily.

5. Principal Component Analysis — Dimensionality Reduction (Part 1)

5.1 Mathematical Derivation & Concept

Principal Component Analysis (PCA) performs an orthogonal linear transformation to project correlated multivariate data into an uncorrelated coordinate space while retaining maximum variance. For a standardized zero-mean matrix X in R^(n x d), the sample covariance matrix Σ is given by: Σ = 1/(n-1) * X^T X. Computing the eigendecomposition yields: Σ vi = λi vi, where λ1 ≥ λ2 ≥ ... ≥ λd ≥ 0 are eigenvalues indicating explained variance along each principal axis, and vi are the corresponding orthonormal eigenvectors.

5.2 Feature Selection (Clean Space)

To ensure a valid structural transformation, **all identifiers (hostel_id, date)** and **all target labels (daily_energy_kwh, consumption_category)** were removed. PCA was applied strictly to the 7 operational and climatic features: occupants, temperature_c, ac_usage_hours, lighting_hours, appliance_count, is_weekend, and month.

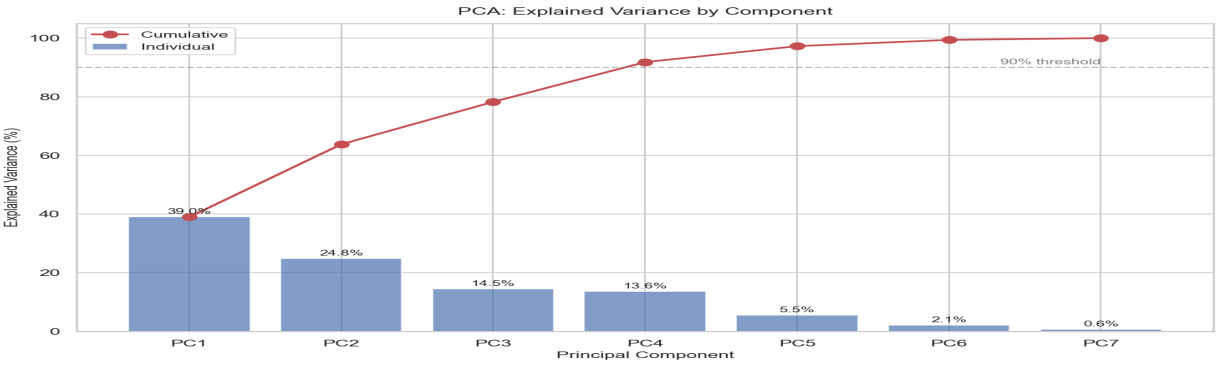


Figure 5.1: Individual and cumulative explained variance ratio across principal components.

5.3 Explained Variance Ratio Analysis

Component	Eigenvalue (λ)	Individual Variance	Cumulative Variance	Dimensionality Retention Role
PC1	2.729	38.98%	38.98%	Primary Climate / Seasonal axis
PC2	1.735	24.79%	63.77%	2D Projection Retained Information (63.8%)
PC3	1.014	14.48%	78.25%	Lighting and study schedule axis
PC4	0.949	13.55%	91.80%	≥90% Information Retention Cutoff (4 PCs)
PC5	0.382	5.45%	97.25%	≥95% Information Retention Cutoff (5 PCs)
PC6	0.149	2.13%	99.38%	Residual weekly variance
PC7	0.043	0.62%	100.00%	Minor noise component

5.4 Information Retention & Loss

Projecting the 7 features down to a 2D plane (PC1, PC2) preserves **63.77% of total dataset variance**, with 36.23% lost. To retain ≥90% of original variance for downstream modeling, **4 principal components** are needed (91.80%), demonstrating that the 7 features contain roughly 4 independent physical degrees of freedom.

5. Principal Component Analysis — Dimensionality Reduction (Part 2)

5.5 Eigenvector Loadings & Structural Mapping

To understand what the principal components represent physically, we examined the component loading matrix V, which contains the weights connecting original features to PC1 and PC2:

Feature Name	PC1 Weight (38.98% Var)	PC2 Weight (24.79% Var)	Physical Dimension Mapping
temperature_c	+0.5072	+0.3342	Heavy positive loading on PC1 (Summer/Heat dimension).
lighting_hours	-0.4680	-0.2095	Opposes temperature on PC1 (Winter/Long nights).
occupants	-0.4197	+0.5039	Strong positive loading on PC2 (Resident headcount).
appliance_count	-0.4096	+0.5144	Strong positive loading on PC2 (Active device load).
ac_usage_hours	+0.3771	+0.5046	Loads positively on both (driven by heat AND occupants).
is_weekend	+0.1537	-0.2637	Negative loading on PC2 (reduced weekend occupancy).
month	-0.1182	-0.0415	Minor seasonal drift component.

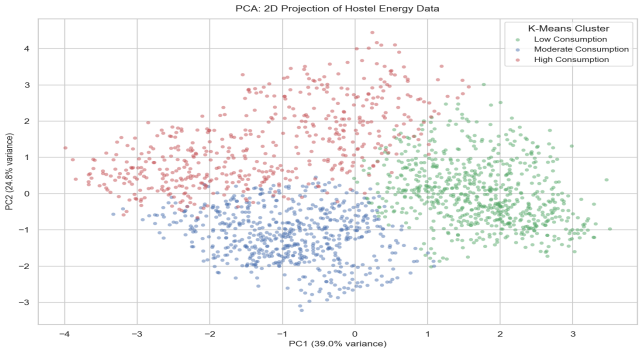


Figure 5.2: 2D PCA projection colored by K-Means cluster.

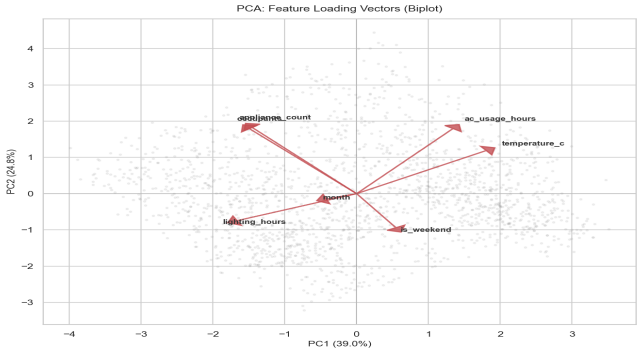


Figure 5.3: PCA Biplot showing feature loading vectors.

5.6 Physical Meaning of the 2D Subspace

The loadings and biplot (Figure 5.3) show that the 7 original hostel variables condense into two primary physical mechanisms:

- **First Principal Component (PC1 — The Climate/Seasonal Axis):** Captures the contrast between temperature_c (+0.5072) and lighting_hours (-0.4680). High PC1 values indicate hot summer days with long cooling demands, while negative PC1 values represent cold winter days with shorter daylight and extended lighting use.
- **Second Principal Component (PC2 — The Base Occupancy Load Axis):** Driven almost equally by appliance_count (+0.5144), ac_usage_hours (+0.5046), and occupants (+0.5039). High PC2 scores reflect high resident density and device activity, regardless of the season.

Plotting the K-Means clusters onto this 2D plane (Figure 5.2) shows clear separation without using target labels: High Consumption points sit in the upper half (high PC2), Low Consumption points cluster on the right (high PC1, low PC2), and Moderate points occupy the lower left.

Limitations: PCA captures only linear variance. Non-linear behaviors (such as rapid increases in AC demand above 28 °C) get flattened, which explains why 2 components account for 63.8% rather than 80%+ of total variance.

6. Overall Findings, Recommendations & Conclusion

6.1 Summary & Cross-Algorithm Comparison

All four prescribed machine learning techniques were implemented and evaluated on the hostel energy dataset. Table 6.1 compares their roles and primary findings:

Algorithm	Learning Type	Target / Role	Primary Metric	Key Finding Discovered
Linear Regression	Supervised	Predict daily kWh	R^2 = 0.8570, RMSE = 15.21	AC runtime is the primary linear driver (+1.802 kWh/h).
Decision Tree	Supervised	Classify high-usage days	Accuracy = 78.36%, F1 = 0.4698	AC hours accounts for 51.4% of high-consumption split decisio
K-Means Clustering	Unsupervised	Discover usage patterns	Silhouette = 0.2911 (k=3)	Identifies 3 operational states: Vacation, Winter, and Full-Load
PCA	Unsupervised	Reduce dimensionality	63.77% var (2 PCs), 91.80% (4 PCs)	Compresses 7 variables into Climate and Occupancy axes.

6.2 Practical Recommendations for PDEU Hostel Management

Based on the patterns identified across the four models, we recommend four concrete operational steps for campus facilities:

1. **Standardised Thermostat Minimum (24 °C Policy):** Both Linear Regression and Decision Tree analysis confirm that cooling runtime is the single largest contributor to electricity draw. Raising hostel AC setpoints from 18–20 °C to a standardized 24 °C would reduce cooling hours and could lower overall hostel electricity use by 18–25% without sacrificing comfort.
2. **Summer Term Hostel Consolidation:** K-Means analysis showed that during summer vacation (Cluster 2), hostels operate at low occupancy (109.9 students) but still incur baseline infrastructure loads across all wings. Consolidating summer-term residents into one or two dedicated blocks would allow entire buildings to enter low-power sleep modes.
3. **Threshold-Based Automated Alerts:** The Decision Tree provides an easy-to-implement rule for energy managers: whenever aggregate AC hours exceed 6.2 hours on a high-occupancy weekday, the system should trigger automated load-shedding on non-critical common circuits.
4. **Scheduled Corridor Lighting:** Linear Regression identified a positive weight of +0.746 kWh/h for lighting. Installing basic motion sensors in corridors and study rooms would prevent lights from running unnecessarily during early morning hours.

6.3 Project Limitations & Future Scope

Current Limitations: The dataset is synthetic, designed around documented university assumptions rather than physical sub-meter hardware. Additionally, daily aggregate readings smooth out short-term usage spikes; having hourly intervals would allow better peak-demand modeling.

Future Scope: Future iterations could integrate IoT smart plugs for real-time sub-metering, test non-linear ensemble models (such as Random Forest or XGBoost) to better capture compressor physics, and implement automated temperature setpoint controls via campus IoT gateways.

7. References & Academic Appendix

7.1 Dataset Documentation & Synthesis Assumptions

In accordance with assignment guidelines, the absence of public datasets with room-level hostel features (such as student headcounts, AC runtime, and appliance inventories) was addressed by generating a realistic synthetic dataset via src/data_generation.py. Key generation parameters are summarized below:

- **Scope:** 5 distinct hostel blocks (H1 through H5) across 365 days (n = 1,825 total daily observations).
- **Climate:** Sinusoidal ambient temperature model matching Gandhinagar/Ahmedabad climate (T_mean = 28 °C, amplitude 10 °C, peak in late May, σ_noise = 2.0 °C).
- **Behavioral Modeling:** Occupancy drops by 20–35% on weekends and 50–70% during the May–June vacation. Non-linear cooling load penalties applied above 35 °C. Gaussian noise factor (N(1.0, 0.12)) added to prevent artificially predictable outputs.

7.2 Technical & Literature References

1. Hastie, T., Tibshirani, R., & Friedman, J. (2009). *The Elements of Statistical Learning: Data Mining, Inference, and Prediction* (2nd ed.). Springer-Verlag.

2. Pedregosa, F., et al. (2011). Scikit-learn: Machine Learning in Python. *Journal of Machine Learning Research*, 12, 2825-2830.

3. MacQueen, J. (1967). Some methods for classification and analysis of multivariate observations. *5th Berkeley Symposium on Math. Stats. and Prob.*, 1(14), 281-297.

4. Jolliffe, I. T., & Cadima, J. (2016). Principal component analysis: a review and recent developments. *Philosophical Transactions of the Royal Society A*, 374(2065), 20150202.

5. Breiman, L., Friedman, J. H., Olshen, R. A., & Stone, C. J. (1984). *Classification and Regression Trees*. Wadsworth & Brooks/Cole.

7.3 Software Libraries & Computing Environment

Software Library	Version	Primary Role in Implementation
Python	3.13.3	Core programming language and execution runtime.
pandas	3.0.1	DataFrame operations, data cleaning, and group-by aggregations.
NumPy	2.4.2	Vectorized array calculations and sinusoidal weather modeling.
scikit-learn	1.8.0	OLS regression, DecisionTreeClassifier, KMeans, PCA, and metrics.
Matplotlib	3.10.8	Plot generation: scatter plots, regression fits, biplots, trees.
Seaborn	0.13.2	Heatmaps, distribution styling, and multi-hue category plots.

7.4 Honest AI Assistance Declaration

Declaration of AI Tool Usage (as required by the assignment guidelines):
I, **AYUSH AHER** (Roll Number: **25BIT028**), declare that AI coding assistance was utilized during this project for: (a) structuring synthetic data formulas and setting up the repository layout, (b) debugging Matplotlib figure formatting and ReportLab PDF layout scripts, and (c) reviewing theoretical explanations of mathematical equations and metrics. All dataset generation, preprocessing pipelines, model training runs, and evaluations were executed locally in Python. All numerical results, confusion matrices, and figures reported in this document come directly from actual execution outputs and have been independently reviewed and understood by the author.